DATA ANALYTICS PROJECT

by

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Alfido Tech Internship

CUSTOMER BEHAVIOR ANALYSIS

Analyzing customer behavior patterns through data-driven insights to support better business decisions.

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Project Report

on

CUSTOMER BEHAVIOR ANALYSIS

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CUSTOMER BEHAVIOR ANALYSISOJECT PLAN

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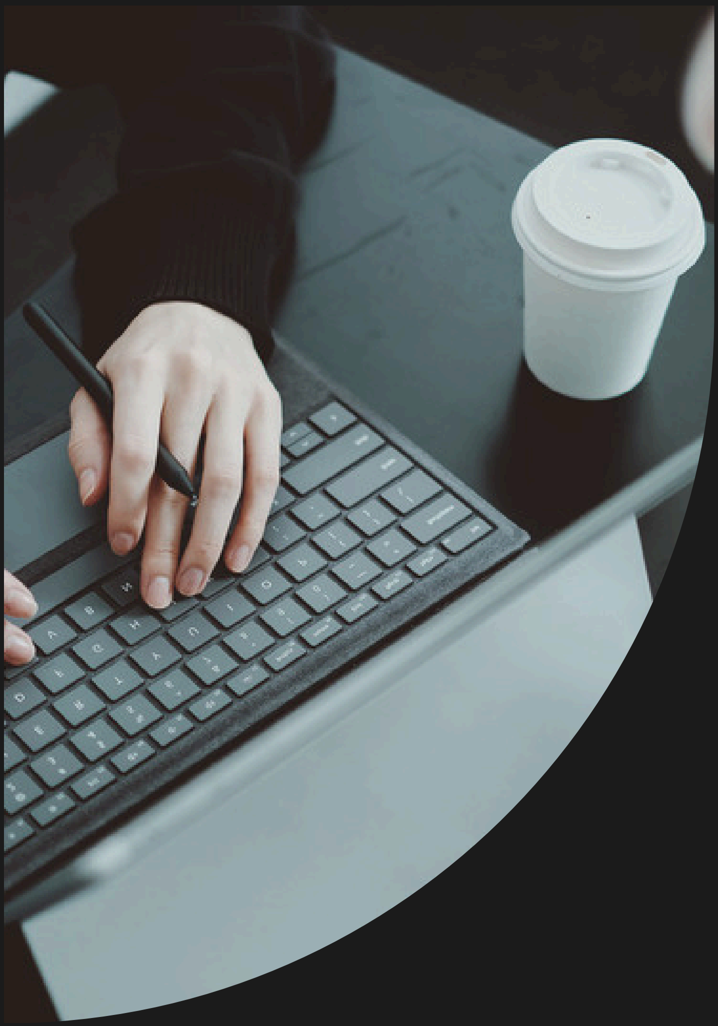
Introduction



Customer behavior analysis is the process of studying how customers interact with products and services and understanding the factors that influence their purchasing decisions. Businesses use customer behavior analysis to identify customer preferences, buying patterns, and trends that can help improve sales and customer satisfaction.

In today's competitive market, organizations collect large amounts of customer data from online and offline transactions. Analyzing this data enables businesses to make data-driven decisions, personalize customer experiences, and develop effective marketing strategies. Customer behavior analysis also helps companies identify high-value customers, understand product demand, and improve customer retention.

The objective of this project is to analyze customer purchasing behavior using an e-commerce dataset. The analysis focuses on customer demographics, purchase patterns, payment methods, product categories, and customer churn. The insights generated from this study can help businesses improve marketing strategies, customer engagement, and overall business performance.

Opening Words

In today's digital world, businesses collect large amounts of customer data. Analyzing this data helps organizations understand customer behavior, improve decision-making, and enhance business performance. This project focuses on Customer Behavior Analysis to identify meaningful patterns and generate valuable business insights.

Objectives:

The primary objective of this Customer Behavior Analysis project is to examine customer purchasing patterns and transform raw transaction data into meaningful business insights.

The specific objectives of this project are:

- **Analyze Customer Purchasing Behavior**
- To study how customers interact with products and services by examining their purchase history, spending patterns, and transaction behavior.
- **Identify Popular Product Categories**
- To determine which product categories receive the highest customer interest and contribute significantly to overall sales performance.
- 2.3 **Understand Customer Demographics**
- To analyze customer characteristics such as age and gender and understand how these factors influence purchasing decisions.
- **Evaluate Customer Retention and Churn**
- To investigate customer churn patterns and identify potential opportunities to improve customer retention and loyalty.
- **Perform Exploratory Data Analysis**
- To explore the dataset using statistical methods and visualizations in order to uncover trends, patterns, and hidden relationships within the data.
- **Generate Actionable Business Insights**
- To provide meaningful recommendations that can help businesses improve marketing strategies, customer engagement, product offerings, and overall business performance.

Dataset Description

The dataset used for this analysis contains customer transaction records from an e-commerce platform. It includes information related to customers, products, purchases, payment methods, and churn status.

Dataset Features

- Customer ID
- Purchase Date
- Product Category
- Product Price
- Quantity
- Total Purchase Amount
- Payment Method
- Customer Age
- Returns
- Customer Name
- Gender
- Churn
-

The dataset consists of 250,000 records and 13 columns, providing sufficient information to understand customer purchasing behavior and identify business trends.

Tools and Technologies Used

The following tools and technologies were used during the project:

Python :

Python was used as the primary programming language for data analysis due to its simplicity and powerful data processing capabilities.

Pandas :

Pandas was used for loading, cleaning, and manipulating the dataset efficiently.

NumPy :

NumPy was used for numerical operations and statistical calculations.

Matplotlib :

Matplotlib was used to create visualizations and charts for better understanding of customer behavior patterns.

Jupyter Notebook :

Jupyter Notebook was used as the development environment for executing code, performing analysis, and documenting findings.

Data Loading and Exploration

The first step involved loading the dataset into Python using Pandas.Basic exploration was performed to understand the structure of the dataset.

The following activities were performed:

- Imported required libraries
- Loaded dataset using Pandas
- Displayed first few records
- Checked data types
- Identified missing values
- Examined dataset dimensions

Dataset loading code :

```
import pandas as pddf = 
pd.read_csv(r"C:\Users\Princy\Desktop\AlfidoTe
ch_projects\ecommerce_customer_data_custom_rat
ios (1).csv")df.head()
```

output:

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name	Age	Gender	Churn
0	46251	2020-09-08 09:38:32	Electronics	12	3	740	Credit Card	37	0.0	Christine Hernandez	37	Male	0
1	46251	2022-03-05 12:56:35	Home	468	4	2739	PayPal	37	0.0	Christine Hernandez	37	Male	0
2	46251	2022-05-23 18:18:01	Home	288	2	3196	PayPal	37	0.0	Christine Hernandez	37	Male	0
3	46251	2020-11-12 13:13:29	Clothing	196	1	3509	PayPal	37	0.0	Christine Hernandez	37	Male	0
4	13593	2020-11-27 17:55:11	Home	449	1	3452	Credit Card	49	0.0	James Grant	49	Female	1

Data Cleaning and Preprocessing

Data cleaning is an important step in data analysis because inaccurate or missing data can affect the quality of results.

The following preprocessing activities were performed:

- Checked for missing values
- Identified null values in the Returns column
- Verified data types
- Ensured consistency of records
- Prepared dataset for visualization and analysis

Output of df.info() and df.isnull().sum() :

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 13 columns):
 #   Column                 Non-Null Count   Dtype
---  ------                 --------------   -----
 0   Customer ID            250000 non-null  int64
 1   Purchase Date          250000 non-null  object
 2   Product Category       250000 non-null  object
 3   Product Price          250000 non-null  int64
 4   Quantity               250000 non-null  int64
 5   Total Purchase Amount  250000 non-null  int64
 6   Payment Method         250000 non-null  object
 7   Customer Age           250000 non-null  int64
 8   Returns                202404 non-null  float64
 9   Customer Name          250000 non-null  object
 10  Age                    250000 non-null  int64
 11  Gender                 250000 non-null  object
 12  Churn                  250000 non-null  int64
dtypes: float64(1), int64(7), object(5)
memory usage: 24.8+ MB
```

```
df.isnull().sum()

Customer ID                  0
Purchase Date                0
Product Category             0
Product Price                0
Quantity                     0
Total Purchase Amount        0
Payment Method               0
Customer Age                 0
Returns                  47596
Customer Name                0
Age                          0
Gender                       0
Churn                        0
dtype: int64
```

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to identify customer purchasing patterns and understand customer behavior.

Product Category Distribution :

Interpretation

The graph shows that Clothing and Books are the most popular product categories, each receiving approximately 75,000 purchases. Electronics and Home products have lower purchase counts, with approximately 50,000 purchases each.
This indicates that customers show a stronger preference for Clothing and Books compared to other product categories. Businesses can use this information to allocate inventory more efficiently and focus promotional campaigns on high-demand categories.

Business Insight

High-performing categories such as Clothing and Books should receive additional marketing attention because they contribute significantly to customer purchases.

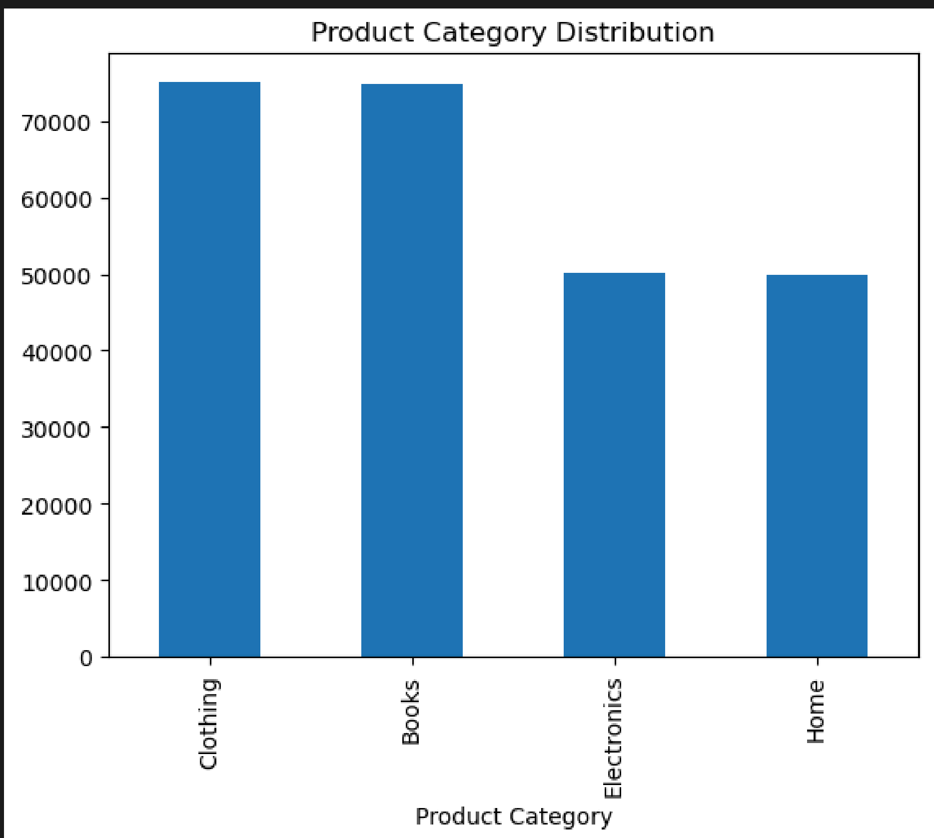


Figure 1: Product Category Distribution Bar Char

Average Spending by Gender:

The second analysis examined whether spending behavior differs between male and female customers.
Insert Graph Here

Interpretation

The graph indicates that average spending by male and female customers is nearly identical. Both genders have an average spending amount of approximately 2,700.
This suggests that gender does not significantly influence customer spending behavior within this dataset.

Business Insight

Since spending patterns are similar across genders, businesses may focus on broader customer segmentation strategies rather than relying solely on gender-based marketing campaigns.

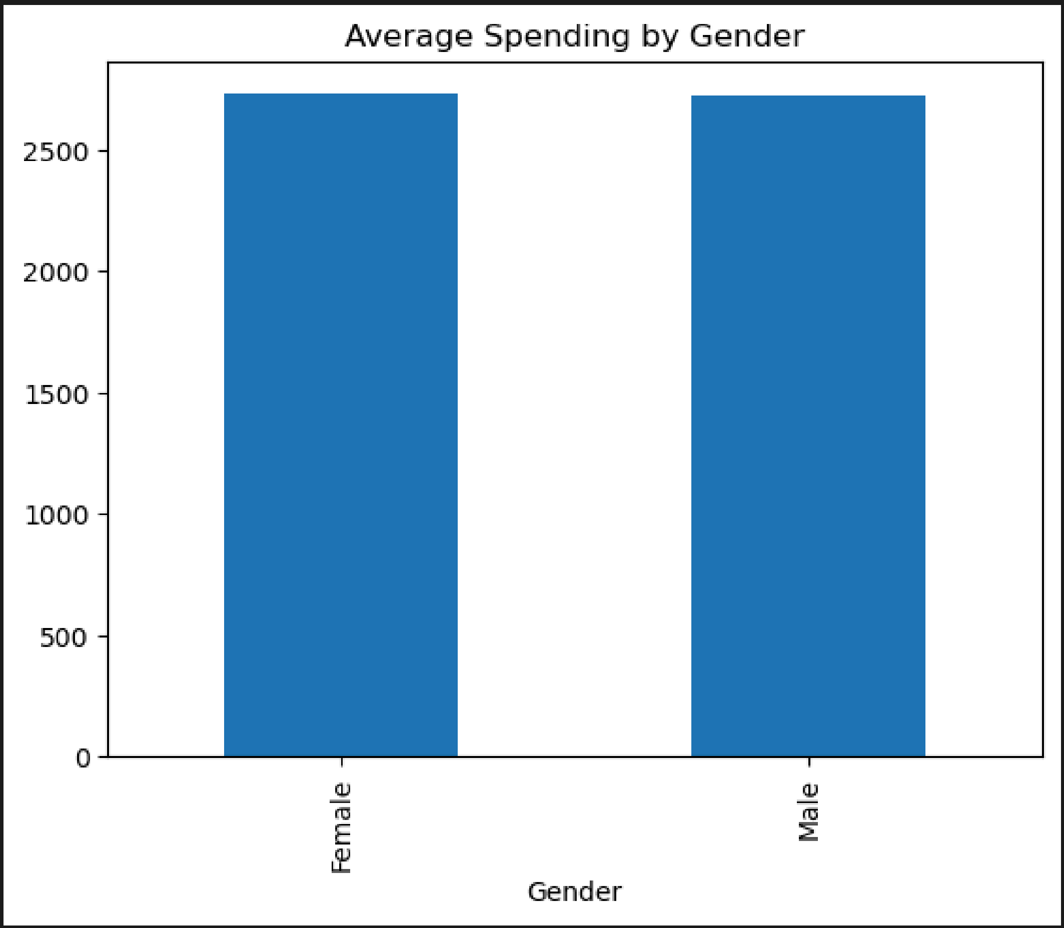


Figure 2: Average Spending by Gendert

Customer Churn Distribution

Customer churn analysis helps identify the percentage of customers who stop engaging with the business.

Interpretation

The pie chart shows that approximately 80.1% of customers remain active, while 19.9% of customers are classified as churned.
Although the majority of customers continue to engage with the business, the churn rate of nearly 20% indicates that a significant portion of customers stop interacting with the platform.

Business Insight

Reducing customer churn can significantly improve long-term profitability. Customer loyalty programs, personalized recommendations, and targeted promotions may help retain customers.

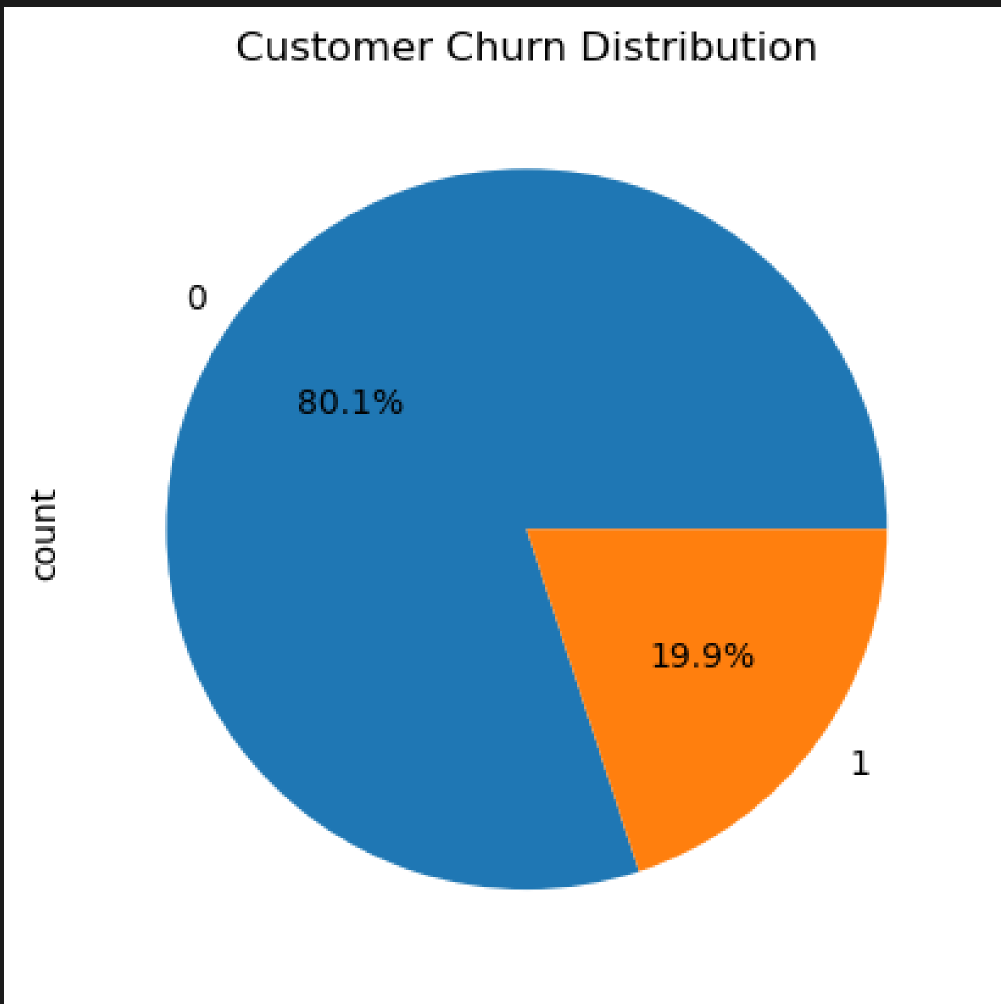


Figure 3: Customer Churn Distribution

Key Findings

The analysis produced several important findings:
Finding 1
Clothing and Books are the most popular product categories among customers.

Finding 2
Male and female customers exhibit very similar spending behavior.

Finding 3
Approximately 20% of customers have churned, indicating opportunities for customer retention improvement.

Finding 4
The majority of customers remain active and continue purchasing products.

Recommendations

Based on the analysis, the following recommendations are proposed:
Recommendation 1 :
Increase marketing efforts for high-performing categories such as Clothing and Books.

Recommendation 2 :
Develop customer loyalty programs to reduce churn and improve retention.

Recommendation 3 :
Implement personalized product recommendations to increase customer engagement.

Recommendation 4 :
Monitor customer behavior regularly to identify early signs of churn.

Conclusion

This project successfully analyzed customer behavior using an e-commerce dataset. Through data exploration, visualization, and interpretation, valuable insights were obtained regarding product category preferences, spending patterns, and customer churn.

The findings indicate that Clothing and Books are the most preferred product categories, spending behavior is similar across genders, and approximately one-fifth of customers have churned.

These insights can help businesses improve customer retention, optimize marketing strategies, and enhance overall business performance.

References

- Python Documentation
- Pandas Documentation
- Matplotlib Documentation
- Kaggle Dataset